

Bandwidth and Propagation Delay Exercises ANS

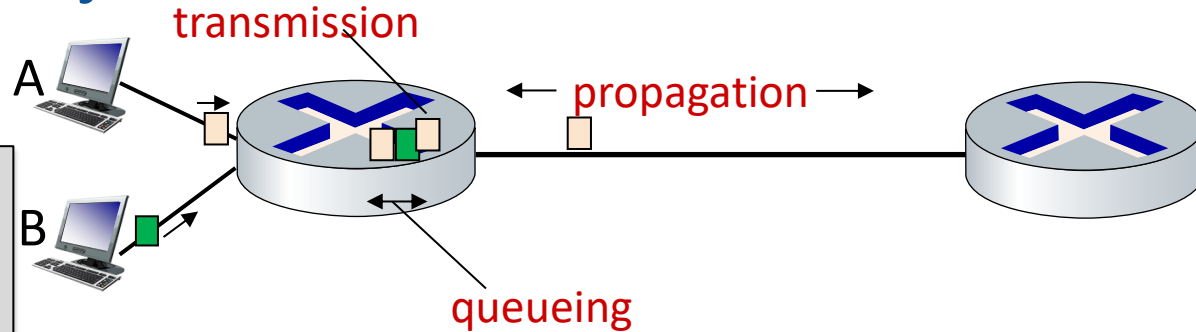
Lecture 3, Spring 2026

Links

- **Bandwidth and Propagation Delay**
- **Pipe Diagrams**
- **Overloaded Links**

Brief Preview of the Semester

Packet delay: four sources



The nodal processing delay (check bit errors, determine output link) is typically very small and can be ignored

$$d_{\text{nodal}} = d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$$

d_q : queueing delay

- time waiting at output link for transmission
- depends on congestion level of router

d_{tx} : transmission delay:

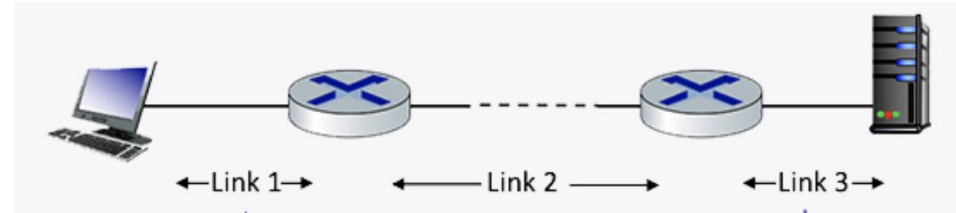
- L : packet length (bits)
- R : link *transmission rate* (bps)
- $d_{\text{trans}} = L/R$

d_{prop} : propagation delay:

- d : length of physical link
- s : propagation speed ($\sim 2 \times 10^8$ m/sec)
- $d_{\text{prop}} = d/s$

Q1 End-to-end delay

- Consider the network shown in the figure, with three links, each with a transmission rate of 1 Mbps, and a propagation delay of 1 msec per link. Assume the length of a packet is 1000 bits.
- What is the end-end delay of a packet from when it first begins transmission on link 1, until it is received in full by the server at the end of link 3. Assume that queueing delays are zero.



Q1 End-to-end delay ANS

- Packet length $L = 1000\text{bits}$, link rate $R = 1\text{Mbps}$ → transmission delay per link
 $t_{tx} = L/R = 1000/1,000,000 = 0.001\text{s} = \mathbf{1\ ms}$.
- Propagation delay per link = **1 ms**.
- Each link contributes $t_{tx} + t_{prop} = 1 + 1 = 2\text{ms}$.
- Three links → total $3 \times 2\text{ms} = \mathbf{6\ ms}$.

Q2 End-to-end delay

- Consider the scenario shown in Figure 1: a server is connected to a router by a 100Mbps link with a 50ms propagation delay. This router is connected to two other routers, each over a 50Mbps link with a 200ms propagation delay. A 1Gbps link connects each host to each of these routers with 0 propagation delay. All packets in the network have size of 20,000 bits.
- What is the end-to-end delay (in ms) from when a packet is transmitted by the server to when it is received by the client? Assume there's no queuing delay at the routers.

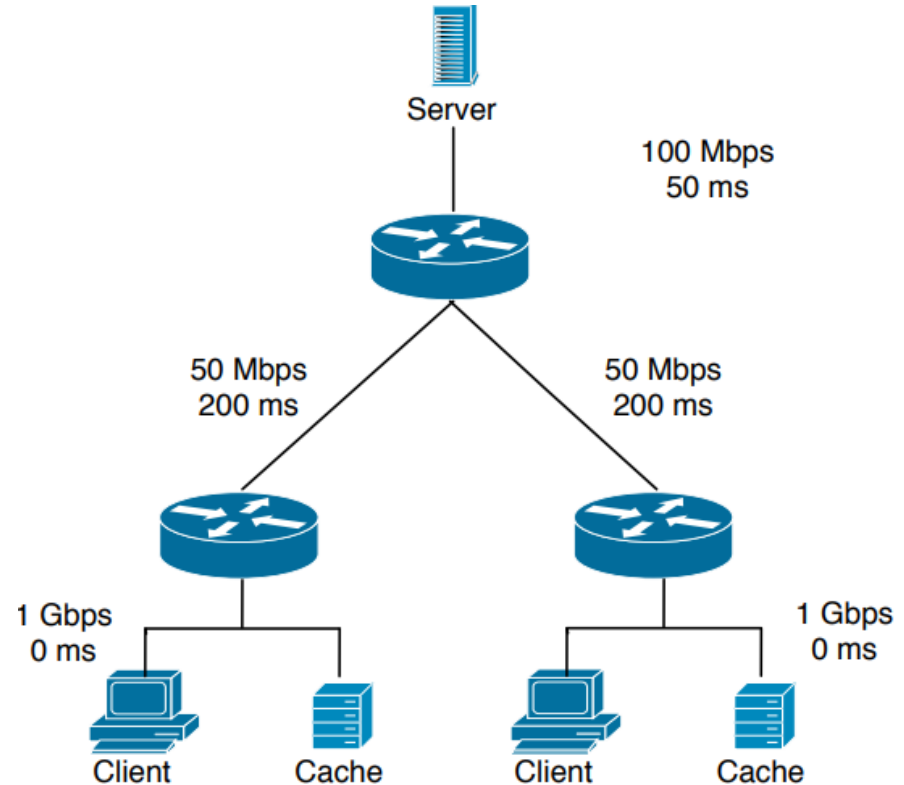


Figure 1

Q1 End-to-end delay ANS

- **Transmission delays:**
- **100 Mbps link:** $20,000/100,000,000 = 0.0002\text{s} = \mathbf{200\ \mu\text{s}}$
- **50 Mbps link:** $20,000/50,000,000 = 0.0004\text{s} = \mathbf{400\ \mu\text{s}}$
- **1 Gbps link:** $20,000/1,000,000,000 = 0.00002\text{s} = \mathbf{20\ \mu\text{s}}$
- Total transmission delay: $t_{tx} = 200 + 400 + 20 = 620\ \mu\text{s} = \mathbf{0.62\ ms}$
- **Propagation delays:**
- Server $\rightarrow$ router: **50 ms**
- Router $\rightarrow$ router: **200 ms**
- Router $\rightarrow$ client: **0 ms**
- Total propagation delay: $t_{prop} = 50 + 200 = 250\ ms$
- **End-to-end delay**
- $250\ ms + 0.62\ ms = 250.62\ ms$

Q3 End-to-end delay

- Consider a 100 Mbps link between an earth station (sender) and a satellite (receiver) at an altitude of 2100 km. The signal propagates at a speed of 3×10^8 m/s. Calculate the time taken (in ms) for the receiver to receive a packet of 1000 bytes transmitted by the sender. (There is no router so there is no queuing delay.)

Q3 End-to-end delay ANS

- **1. Transmission delay.**
- Packet size: 1000bytes = 8000bits
- Link rate: 100Mbps = 100×10^6 bits/s
- $t_{tx} = \frac{8000}{100 \times 10^6} = 8 \times 10^{-5} \text{ s} = 0.08 \text{ ms}$
- **2. Propagation delay**
- Distance: 2100km = 2.1×10^6 m
- Propagation speed: 3×10^8 m/s
- $t_{prop} = \frac{2.1 \times 10^6}{3 \times 10^8} = 7 \times 10^{-3} \text{ s} = 7.00 \text{ ms}$
- **3. Total delay**
- $t_{total} = t_{tx} + t_{prop} = 0.08 + 7.00 = 7.08 \text{ ms}$

Q4 End-to-end delay

- Consider two computers X and Y connected via a single link of bandwidth 512 Gbps. Suppose that both hosts are separated by distance M meters, and the propagation delay along the link is 2×10^9 meters/sec. Computer X has to send a packet of size 1 Kbyte to computer Y. What will be the distance M such that the propagation delay is equal to the transmission delay?

Q4 End-to-end delay ANS

- **Step 1: Transmission delay**

- $t_{tx} = \frac{\text{packet size}}{\text{bandwidth}} = \frac{8192}{512 \times 10^9} = \frac{16}{10^9} = 1.6 \times 10^{-8} \text{ s}$

- (1 Kbyte = 1024 bytes = 1024×8 = 8192 bits. Notice it is NOT 1000 bytes.)\

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- **Step 2: Propagation delay**

- $t_{prop} = \frac{M}{2 \times 10^9}$

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- **Step 3: Set them equal**

- $\frac{M}{2 \times 10^9} = 1.6 \times 10^{-8}$

- $M = 32 \text{ meters}$

Q5 Transmission delay

- 1. Calculate transmission delay for message size $L = 8000\text{bits}$, network bandwidth $B = 8000\text{bits/sec}$
- ANS:
- 2. Calculate transmission delay for message size $L = 1\text{ Kbytes}$, network bandwidth $B = 8\text{ Kbps}$
- ANS:

Q5 Transmission delay ANS

- 1. Calculate transmission delay for message size $L = 8000\text{bits}$, network bandwidth $B = 8000\text{bits/sec}$
- ANS: $t_{tx} = \frac{L}{B} = \frac{8000 \text{ bits}}{8000 \text{ bits/sec}} = 1 \text{ sec}$
- 2. Calculate transmission delay for message size $L = 1 \text{ Kbytes}$, network bandwidth $B = 8 \text{ Kbps}$
- ANS: $t_{tx} = \frac{L}{B} = \frac{1 \text{ Kbytes}}{8 \text{ Kbps}} = \frac{8192 \text{ bits}}{8000 \text{ bits/sec}} \approx 1.02 \text{ sec}$